

Question ID: 3148fe3e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$x^2 + y + 10 = 10$$
$$8x + 16 - y = 0$$

The solution to the given system of equations is (x, y) . What is the value of x ?

Answer

- A. -16
- B. -4
- C. 2
- D. 8

Correct Answer: B

Rationale

Choice B is correct. Adding y to each side of the second equation in the given system of equations yields $8x + 16 = y$. Substituting $8x + 16$ for y in the first equation yields $x^2 + 8x + 16 + 10 = 10$. Subtracting 10 from each side of this equation yields $x^2 + 8x + 16 = 0$. This equation can be rewritten as $(x + 4)^2 = 0$. Taking the square root of each side of this equation yields $x + 4 = 0$. Subtracting 4 from each side of this equation yields $x = -4$. Therefore, the value of x is -4 .

Choice A is incorrect. This is the value of y , not x .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.